Chapter 4. Decision Trees and Random Forests

Predictive Modeling, Lecture 3 (CART, impurity, pruning, bagging, OOB, random forest, importance)

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1 After the Linear Baseline: Where Does the Relation Change?

Linear models need hand-designed interactions; trees split the input space by if–then conditions with a constant per region. **A tree discovers interactions and thresholds; a random forest stabilizes it.**

2 The Tree as a Piecewise-Constant Function

Motivation. Rectangles with constants give interactions for free but high variance.

Definition: regression tree

$$\hat{f}(x) = \sum_{m=1}^{M} c_m \, \mathbb{I}(x \in R_m),$$

where $R_1, \ldots, R_M$ are the leaf regions (a partition of $\mathcal{X}$), c_m is the prediction in region m, and the boundaries are axis-aligned splits by a threshold on one variable.

Each leaf is a rectangle and each root-to-leaf path an if–then rule. One changed split changes everything below it: high variance.

2.1 Nested conditions create interactions automatically

Two splits already give

$$\hat{f}(x) = \begin{cases} c_1, & x_1 < s_1, \\ c_2, & x_1 \geq s_1 \ \wedge \ x_2 < s_2, \\ c_3, & x_1 \geq s_1 \ \wedge \ x_2 \geq s_2. \end{cases}$$

x_2 matters only when $x_1 \geq s_1$: an interaction, which a linear model needs as a hand-made $x_1 x_2$.

2.2 Leaf predictions

Theorem: optimal leaf constants

For a fixed partition, (i) under squared loss the optimal c_m is the leaf mean $\hat{c}_m = \frac{1}{N_m} \sum_{i: x_i \in R_m} y_i$;
(ii) under absolute loss it is the leaf median;
(iii) for classification, the leaf class proportions $\hat{p}_{mk} = \frac{1}{N_m} \sum_{i: x_i \in R_m} \mathbb{I}(y_i = k)$ minimize the log loss, and $\hat{y}(x) = \arg\max_k \hat{p}_{mk}$ minimizes misclassification.

Proof [Supplement]. The loss separates over leaves; in each, apply the constant-prediction result of Ch. 1 to the leaf's empirical distribution. ■

Predictions jump like a staircase at leaf boundaries. $\hat{p}_{mk}$ is a probability estimate, unstable in small leaves; check calibration before acting.

3 Growing the Tree: CART

3.1 One variable, one threshold per node

$$R_1(j, s) = \{x : x_j < s\}, \qquad R_2(j, s) = \{x : x_j \geq s\}.$$

Search all (j, s), take the best split, recurse: **greedy**, not globally optimal.

3.2 Regression split criterion

$$\Delta_{\text{RSS}}(j,s) = \sum_{i \in R} (y_i - \bar{y}_R)^2 - \sum_{r \in \{1,2\}} \sum_{i \in R_r(j,s)} (y_i - \bar{y}_{R_r})^2.$$

Choose (j, s) maximizing Δ_{RSS}.

Derivation: efficient search and the ANOVA form of the gain

For node R with $N = N_1 + N_2$, the within/between decomposition gives

$$\Delta_{\text{RSS}} = N_1(\bar{y}_{R_1} - \bar{y}_R)^2 + N_2(\bar{y}_{R_2} - \bar{y}_R)^2 = \frac{N_1 N_2}{N}(\bar{y}_{R_1} - \bar{y}_{R_2})^2 \geq 0.$$

Continuous x_j: after sorting, running sums update in $O(1)$ per cut, $O(N \log N)$ in total. Categorical with K levels: $2^{K-1} - 1$ partitions, but sorting levels by mean target leaves $K - 1$ (Exercise 4.1).

3.3 Classification split criterion

$$G(R) = 1 - \sum_{k=1}^{K} p_{Rk}^2 \quad \text{(Gini)}, \qquad H(R) = -\sum_{k=1}^{K} p_{Rk} \log p_{Rk} \quad \text{(entropy)}.$$

Pure leaves have impurity 0; minimize $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2)$. Gini and entropy give similar trees.

Theorem: interpretation and monotonicity of impurity

(i) $G(R)$ is the probability that two labels drawn independently from the leaf distribution differ, equivalently the misclassification rate of a rule that labels at random according to p_R. (ii) Gini and entropy are strictly concave functions of p; hence any split satisfies $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2) \leq I(R)$, with equality iff both children have the parent's class proportions.

Proof [Supplement]. (i) $P(\text{two draws differ}) = 1 - \sum_k p_k^2$.
(ii) $p_R = \frac{N_1}{N} p_{R_1} + \frac{N_2}{N} p_{R_2}$; apply Jensen to the concave impurity. ■

$1 - \max_k p_k$ is only piecewise linear, so many splits gain zero (Exercise 4.2). The split criterion is not the evaluation metric.

3.4 Many candidate splits cause selection bias

Variables with many cut points or levels get more chances to split, and the best training split in a small node may not generalize. Remedies: minimum leaf size, pruning, validation; check high-cardinality IDs for leakage.

3.5 Greedy CART: fast but not globally optimal

Early splits are never revisited and lock in later structure; a tree's structure is not "the discovered truth".

3.6 Axis-aligned splits approximate oblique boundaries by staircases

Threshold rules suit trees; $x_1 + x_2 > c$ or smooth functions need many small, high-variance leaves. Compare with linear baselines under one protocol.

4 Controlling Complexity

Depth always lowers training error; test error can rise again.

Approach	Example	Pros and cons
pre-pruning	maximum depth, minimum leaf size, minimum split gain	fast, but stopping too early can miss good structure
post-pruning	grow large, then remove branches	sees a wider set of candidates, but requires a separate complexity choice
validation-based choice	choose depth / leaf size / pruning strength by CV	targets generalization; must be done inside the pipeline

Choose depth by generalization performance.

4.1 Minimum leaf size is a statistical regularizer

$$\hat{p}_m = \frac{n_{m,+}}{N_m}, \qquad \mathrm{SE}(\hat{p}_m) \approx \sqrt{\frac{\hat{p}_m(1-\hat{p}_m)}{N_m}}.$$

Small leaves are noisy ($N_m = 5$, $\hat{p} = 0.4$: SE ≈ 0.22); large leaves mix groups (bias). Minimum leaf size is a regularizer, not a speed option.

4.2 Cost-complexity pruning

$$R_\alpha(T) = R(T) + \alpha|T|,$$

where $R(T)$ is the training risk, $|T|$ the number of leaves and α the price per leaf, chosen by CV.

Derivation: weakest-link pruning

For an internal node t with subtree T_t, collapsing T_t to a leaf changes R_α by $\{R(t) - R(T_t)\} - \alpha(|T_t| - 1)$. Collapsing pays off as soon as

$$\alpha \geq g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1}.$$

Collapsing the smallest-$g(t)$ node repeatedly gives nested $T_0 \supset T_1 \supset \cdots \supset \{\text{root}\}$ at $\alpha_1 < \alpha_2 < \cdots$, containing the optimal subtree for every α (Breiman et al.); CV chooses among finitely many trees.

5 From One Tree to Many: Bagging

5.1 A single tree has high variance

A few changed observations can flip the root split and everything below. Averaging slightly different trees cancels this variance: **trees are strong but unstable**.

5.2 Bagging

Algorithm: bagging (bootstrap aggregating)

For $b = 1, \dots, B$: draw a bootstrap sample of size n with replacement; grow a deep tree $\hat{f}_b$. Aggregate by averaging (regression) or by averaging probabilities / majority vote (classification).

A single tree is pruned; ensemble trees are grown deep (low bias) and averaging lowers variance, though leaf size and depth still matter for small samples, rare classes and probabilities.

5.3 A bootstrap sample contains about 63.2% of the distinct observations

Theorem: inclusion probability

The probability that a given observation never appears in a bootstrap sample of size n is $(1 - 1/n)^n \to e^{-1} \approx 0.368$. Hence on average about $1 - 0.368 = 0.632$ of the distinct observations are used to grow one tree.

Proof [Supplement]. Each of n independent draws misses it w.p. $1 - 1/n$, and $n \log(1 - 1/n) = -1 - \frac{1}{2n} + O(n^{-2}) \to -1$. ■

Left-out (out-of-bag, OOB) observations allow internal evaluation.

5.4 Averages are sensitive to correlation between trees

Theorem: variance of an average of correlated predictors

If each $\hat{f}_b(x)$ has variance σ^2 and every pair has correlation $\rho \geq 0$, then

$$\operatorname{Var}\Big(\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)\Big) = \sigma^2\Big[\rho + \frac{1-\rho}{B}\Big].$$

Proof [Supplement]. $\operatorname{Var}(\sum_b \hat{f}_b) = B\sigma^2 + B(B-1)\rho\sigma^2$. Divide by B^2: $\frac{\sigma^2}{B} + \frac{B-1}{B}\rho\sigma^2 = \sigma^2\{\rho + (1-\rho)/B\}$. ■

The floor is $\rho\sigma^2$; bias is unchanged ($\mathbb{E}[\frac{1}{B}\sum_b \hat{f}_b] = \mathbb{E}\hat{f}_1$). Random forests lower ρ.

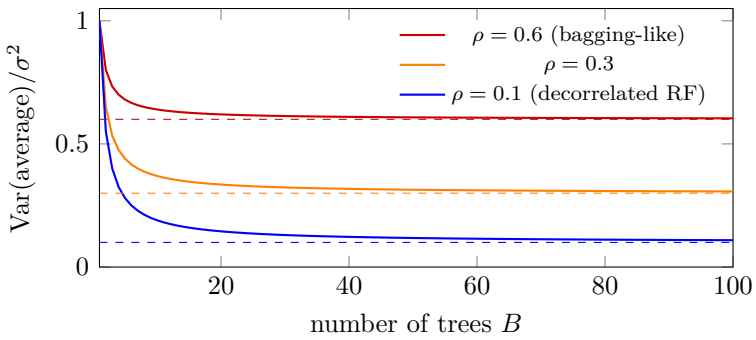


Figure 1: $\rho + (1-\rho)/B$ levels off at the floor ρ (dashed); decorrelation lowers the floor.

Task	Output of one tree	Ensemble output
regression	$\hat{f}_b(x)$	$\frac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)$
binary classification	$\hat{p}_b(x)$ or class	average the probabilities then threshold, or majority vote
multiclass	class probability vector	argmax of the averaged class probabilities
ranking / policy	risk score, probability	combine with top-k, a threshold, a cost rule

6 Random Forests

6.1 Bagging alone can let every tree use the same strong variable

If every tree splits first on the same strong variable, trees are alike and high ρ limits the gain for any B.

6.2 Randomness over both observations and features

Algorithm: random forest

For $b = 1, \ldots, B$: draw a bootstrap sample; grow a tree in which, **at each node**, only m_{try} randomly chosen variables are allowed as split candidates; grow deep (minimum leaf size small). Aggregate as in bagging.

Bootstrap varies the data; feature subsampling varies the structure, lowering ρ.

6.3 m_{try}

$m_{\text{try}} = p$ is bagging; smaller means lower correlation but weaker trees. Start at $\sqrt{p}$ (classification) or $p/3$ (regression); tune by OOB or CV.

6.4 OOB scores are convenient but do not always replace CV

For observation i, average only the trees for which i was out of bag: $\hat{f}^{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|}\sum_{b\in\mathcal{B}_i}\hat{f}_b(x_i)$, with $|\mathcal{B}_i| \approx 0.368B$; the OOB error is the average loss of these predictions.

Method	Strength	Caveat
OOB	fast internal estimate without a separate validation set	does not respect time/group structure; not independent validation
CV	compares preprocessing and candidates under the same rule	expensive; the pipeline must be repeated inside folds
final test	reports generalization of a fixed procedure	used once; not material for re-tuning

OOB does not replace the time/group/leakage rules of Ch. 1.

7 Variable Importance Is Not Causal Importance

Impurity-decrease importance: total impurity reduction from splits on a variable. **Permutation importance**: drop in validation/OOB score after shuffling one column:

$$I_j = \text{Score}(D_{\text{eval}}) - \text{Score}(D_{\text{eval}}^{\pi(j)}),$$

with only column j permuted (sign chosen so a loss increase is positive); use OOB/validation data, not the test set, and repeat permutations.

Importance means the model *relied on* a variable, not that changing it changes the outcome. Correlated substitutes mask each other; impurity importance favors high-cardinality variables.

8 Comparison and Summary

Model	Randomness	Main strength	Main limitation
single tree	none or limited	rules are directly readable	high variance, unstable boundaries
bagging	bootstrap of rows	reduces variance of deep trees	gain limited if trees are correlated
random forest	rows + feature subsets	lower correlation, more stable prediction	hard to explain by one rule
boosting (next)	sequential residual / weight updates	reduces bias, strong prediction	more sensitive to tuning, overfitting, time structure

What matters is **how dissimilar the trees' errors are**.

Summary

1. Trees split the input space by conditions and capture nonlinearity and interactions.
2. A deep single tree is expressive but has high variance and overfits.
3. Bagging averages bootstrap trees to lower variance; random forests add feature subsampling to lower correlation between trees.
4. OOB, importance and partial dependence are useful diagnostics, but do not replace independent validation or causal interpretation.
5. Trees must be compared with the linear baseline under the same split, CV and final-test rules.

9 Selected exercises

Motivation. Level sorting, misclassification as a split criterion, and when aggregation helps.

Exercise 4.1 (Ordering the levels of a categorical variable)

A node R contains N observations of a categorical feature with levels $1, \dots, K$, level k having $N_k \geq 1$ observations with mean target $\bar{y}_k$. A binary split sends a nonempty proper subset L of the levels left and the rest right. Show that some split maximizing Δ_{RSS} has the form $L = \{k : \bar{y}_k \leq \tau\}$ for a threshold τ, so only the $K-1$ splits along the sorted level means need to be examined. Show also that for $y_i \in \{0, 1\}$ the Gini decrease equals $\frac{2}{N}\Delta_{\text{RSS}}$, so the same holds for binary classification with Gini.

Solution [Supplement].

Step 1. With $W(L)$ the within-children SS, $\Delta_{\text{RSS}}(L) = \text{SS}_{\text{tot}} - W(L)$; minimize W. Means minimize SS, so

$$W(L) = \min_{c_1, c_2} C(L, c_1, c_2), \qquad C(L, c_1, c_2) = \sum_{i:\, \text{lev}(i) \in L} (y_i - c_1)^2 + \sum_{i:\, \text{lev}(i) \notin L} (y_i - c_2)^2.$$

Step 2. Within level k, $\sum_{i:\,\mathrm{lev}(i)=k}(y_i - c)^2 = Q_k + N_k(\bar{y}_k - c)^2$, where Q_k does not depend on c.

Step 3. Let $L^\star$ be optimal with child means $c_1 \le c_2$ (else take the complement). If $c_1 = c_2$, $\Delta_{\mathrm{RSS}}(L^\star) = 0 \le \Delta_{\mathrm{RSS}}$ of any split, so all are optimal. Else put $\tau = \frac{1}{2}(c_1 + c_2)$, $L' = \{k : \bar{y}_k \le \tau\}$; by Step 2 level k is cheaper on the c_1 side iff $(\bar{y}_k - c_1)^2 \le (\bar{y}_k - c_2)^2$ iff $\bar{y}_k \le \tau$. Hence

$$W(L') \le C(L', c_1, c_2) \le C(L^\star, c_1, c_2) = W(L^\star).$$

Step 4. L' is proper: otherwise $C(L', c_1, c_2)$ is an SS about one constant, $\ge \mathrm{SS}_{\mathrm{tot}}$, so $\Delta_{\mathrm{RSS}}(L^\star) \le 0$, contradicting $\Delta_{\mathrm{RSS}}(L^\star) = \frac{N_1 N_2}{N}(c_1 - c_2)^2 > 0$. So L' is optimal.

Step 5. For $y_i \in \{0, 1\}$, $G = 2p(1-p)$ and $\mathrm{SS} = Np(1-p)$, so $N\,G(R) = 2\,\mathrm{SS}(R)$ for every node and

$$G(R) - \frac{N_1}{N}G(R_1) - \frac{N_2}{N}G(R_2) = \frac{2}{N}\{\mathrm{SS}(R) - \mathrm{SS}(R_1) - \mathrm{SS}(R_2)\} = \frac{2}{N}\Delta_{\mathrm{RSS}}.$$

■

Exercise 4.2 (Misclassification error ignores useful splits)

Let $E(R) = 1 - \max_k p_{Rk}$ be the misclassification impurity.
(a) Show that if a class $k^\star$ is a majority class of the parent and of both children, then $N_1 E(R_1) + N_2 E(R_2) = N\,E(R)$: the split has zero gain.
(b) A node has 400 cases of each of two classes. Split A gives children $(300, 100)$ and $(100, 300)$; split B gives $(200, 400)$ and $(200, 0)$. Compare the weighted impurities of the two splits under E and under Gini.

Solution [Supplement].

(a) $N\,E(R) = N - n_{k^\star}$, $N_r E(R_r) = N_r - n_{r,k^\star}$, and $N_1 + N_2 - (n_{1,k^\star} + n_{2,k^\star}) = N - n_{k^\star}$.

(b)

Step 1. Under E: A makes $100 + 100$ errors, B $200 + 0$; both $200/800 = 0.25$ (parent 0.5).

Step 2. Gini, A: children $(\frac{3}{4}, \frac{1}{4})$, $G = 1 - \frac{9}{16} - \frac{1}{16} = 0.375$. B: $(\frac{1}{3}, \frac{2}{3})$ with $G = 1 - \frac{1}{9} - \frac{4}{9} = \frac{4}{9}$, weight $\frac{600}{800}$, plus a pure child: $\frac{3}{4} \cdot \frac{4}{9} = \frac{1}{3} < 0.375$.

Gini (and entropy: 0.562 vs. 0.477 nats) prefers B with its pure leaf; E cannot tell. ■

Exercise 4.3 (Aggregation helps under squared loss, not always under a vote)

Let $\varphi(x, \mathcal{D})$ be a predictor fitted on a random training sample $\mathcal{D}$, and $\varphi_A(x) = \mathbb{E}_{\mathcal{D}}\,\varphi(x, \mathcal{D})$ the ideal aggregated predictor (assume $\mathbb{E}_{\mathcal{D}}\varphi(x, \mathcal{D})^2 < \infty$).
(a) Show that for every (x, y), $\mathbb{E}_{\mathcal{D}}\{y - \varphi(x, \mathcal{D})\}^2 = \{y - \varphi_A(x)\}^2 + \mathrm{Var}_{\mathcal{D}}\varphi(x, \mathcal{D})$, and conclude that for a test pair (X, Y) independent of $\mathcal{D}$ the risk of φ_A is at most the expected risk of φ.
(b) For classification let $\varphi(x, \mathcal{D}) \in \{0, 1\}$ and $\varphi_A(x) = \mathbb{I}\{P_{\mathcal{D}}(\varphi(x, \mathcal{D}) = 1) > \frac{1}{2}\}$ (the majority vote over training samples). Give a point x where the vote has a larger error than φ.

Solution [Supplement].

(a)

Step 1. $y - \varphi = (y - \varphi_A) + (\varphi_A - \varphi)$; the cross term $2(y - \varphi_A)\mathbb{E}_{\mathcal{D}}(\varphi_A - \varphi) = 0$.

Step 2. $(X, Y) \perp \mathcal{D}$, so by Fubini

$$\mathbb{E}_{X,Y,\mathcal{D}}\{Y - \varphi(X, \mathcal{D})\}^2 = \mathbb{E}_{X,Y}\{Y - \varphi_A(X)\}^2 + \mathbb{E}_X \mathrm{Var}_{\mathcal{D}}\varphi(X, \mathcal{D}) \;\ge\; \mathbb{E}_{X,Y}\{Y - \varphi_A(X)\}^2.$$

The gain is the variance of the base learner, largest for unstable learners.

(b) $P(Y = 1 \mid x) = 1$, $P_{\mathcal{D}}(\varphi(x, \mathcal{D}) = 1) = 0.4$: φ errs w.p. 0.6, but $\varphi_A(x) = 0$ always errs. A vote amplifies the more frequent answer. ■